

Hannah Kim

714-316-9974 | hannahlkim2006@gmail.com | [linkedin.com/in/hannah-l-kim](https://www.linkedin.com/in/hannah-l-kim) | github.com/saltannah

EDUCATION

University of California, Irvine - Henry Samueli School of Engineering

Bachelor of Science in Mechanical Engineering

Irvine, CA

June 2028

EXPERIENCE

Johnson & Johnson MedTech

June 2026 – Present

R&D Engineering Intern

Irvine, CA

- Develop and validate custom fixtures and test methods for elastomeric sealing components; conducted Gage R&R studies to verify measurement system capability, achieving **22.2% measurement variation** relative to tolerance
- Designed and prototyped 3D printed test fixtures in Solidworks to support sample fabrication and characterization
- Researched existing device architectures and performance requirements to define design factors; Review product requirements and applicable standards (**ISO 11070**) to establish verification criteria and improve test methodologies

UC Irvine Rocket Project Solids

July 2025 – Present

Payload Structures Engineer

Irvine, CA

- Modified CubeSat housing design to **reduce back panel mass by 38%** while maintaining structural integrity and FEA results for a **FoS of 3**; evaluate **friction, tolerances, and material behavior** across PETG, TPU, and metal hardware
- Designed a passive rover deployment mechanism for a 5U CubeSat structure using wheel motor actuation to automatically release a threaded restraint upon landing; iteratively tested 3D-printed components to evaluate release reliability
- Achieved **Level 1** high-power rocketry certification by independently designing, simulating (OpenRocket), and launching a rocket to an estimated apogee of **2700 ft**

Theta Tau Professional Engineering Fraternity

September 2025 – Present

Director of Engineering

Irvine, CA

- **Led an organization-wide engineering initiative** and mentored 20 students in quarter-long multidisciplinary projects, conducting design reviews and providing guidance from concept development through final demonstration
- Coordinated logistics for a regional engineering conference hosting **50+ student representatives from 14 universities**; outreach for technical events, networking opportunities, and professional development programs on campus

ZOTbotics Introductory Makerspace

August 2024 – May 2026

Makerspace Director

Irvine, CA

- Taught introductory hands-on workshops for **3D printing, laser cutting, CNC machining, and power tool usage**
- Directed recruitment and training of **10+ makerspace managers**; oversee operations for **170+ active members**
- Designed and implemented makerspace workflow systems, including a **3D printer reservation system used by 70+ students**, integrating form intake, scripted scheduling to a shared calendar, and automated user notifications

PROJECTS

Autonomous Recruitment Robot | *Mechanical Engineer*

November 2024 – June 2025

- Designed and prototyped a spring-loaded flyer dispensing mechanism in CAD to increase engagement at recruitment events
- Iterated on structural and aesthetic design by analyzing spring behavior, validating tolerances, and optimizing 3D print orientation and parameters for manufacturability and chassis integration

TimelyRx - Smart Medication Dosage Device | *Personal Project - UCI BitHacks*

April 2025

- Integrated SQLite database logic to log and track dosages and created real-time display updates for medication adherence
- Designed a 3D-printed lock mechanism with integrated toggle switch and OLED display, optimizing user accessibility

Self-Navigating Rover | *Software Team Lead*

September 2024 – December 2024

- Programmed an Arduino-based line-following rover, developing functions for IR sensor line tracking, PixyCam object detection, and servo/motor control for autonomous navigation and color-detected object retrieval
- Integrated a PID controller to minimize overshooting and correct path deviations, improving tracking accuracy

Beckman Robotics | *Project Lead*

August 2020 – May 2024

- Focused on chassis design for peak drive performance, optimized lift linkage ratio to increase output torque by 66.7%
- Created custom CAD/CAM models for 3D-printed and CNC-machined parts; customized commercial off-the-shelf (COTS) components through machining to support project-specific applications

SKILLS

Manufacturing/Fabrication: GD&T, 3D Printing, Laser/Plasma Cutting, CNC Machining, Design for Manufacturing

Software: Solidworks, CATIA, Fusion 360, Autodesk Inventor, Java, Python, C/C++, SQLite, MATLAB, Arduino IDE

Coursework: Thermodynamics, Statics, Dynamics, Computer-Aided Design (CAD), Finite Element Analysis (FEA), Circuits